

Frozen Hub Caption Transfer

How a caption-retrieval head trained only from DINOv2-small can be evaluated on ConvNeXt, with the correct train/evaluation split and a cat/dog walkthrough.

Core claim: ConvNeXt does not pass through DINOv2-small at test time. ConvNeXt uses its own fitted entry map into the frozen hub; the frozen caption head was historically trained using DINOv2-small hub coordinates.

Executive Summary

The project builds a shared 512-dimensional hub from several original encoders. A linear caption head is then fitted with **only DINOv2-small-derived hub coordinates** as inputs and real BGE-M3 caption embeddings as targets. When ConvNeXt later arrives, it receives only an independent linear entry map into that existing hub. The DINO-trained head is frozen and applied unchanged.

The reported **96.7%** is a relative performance ratio, not an R@1 score by itself:

$R@1_{\text{transferred}} / R@1_{\text{native ConvNeXt}} = 0.967$. The transferred DINO-small-trained head retains 96.7% of the R@1 achieved by a separate linear head trained specifically on ConvNeXt hub coordinates.

Dataset Split: The Important Correction

There are **9,533 paired image-caption items in total**. That does **not** mean all 9,533 should be used to fit the caption head and then scored again. The stated retrieval evaluation uses 1,000 held-out items; therefore the natural fitting partition is 8,533 items.

Partition	Rows	Permitted role in the strict evaluation
Fitting / train	8,533	Fit ridge maps and the caption heads; choose regularization without using final test R@1.
Held-out retrieval test	1,000	Compute predictions, rank against the held-out BGE gallery, and report R@1. Do not use these rows to fit W_{head} , W_{conv} , or W_{native} .
Total	9,533	All paired items available before the split.

Protocol check - and the project passes it. This is exactly the right question, and it is already answered on the record: the hub basis was estimated on the 8,533 TRAINING rows only, never on all 9,533. The report states it directly - a whitened PCA basis of the concatenated spaces 'estimated on 8,533 training rows', with '1,000 rows held out for every measurement'. The cached artefacts agree: `hub_exact.npz` stores `hub_train` with shape (8533, 512) alongside separate `train_idx` and `eval_idx` arrays. Every entry map and every head is fitted on the same 8,533 rows, and the 1,000 test rows are only ever transformed. The protocol is therefore NOT transductive and needs no disclosure of that kind.

Notation And Models

Symbol	Shape	Meaning
X_small	N x 768	Raw DINOv2-small image vectors.
X_conv	N x 1024	Raw ConvNeXt-base image vectors.
H	N x 512	Frozen shared hub coordinates.
H_small	N x 512	DINOv2-small mapped into the hub.
H_conv	N x 512	ConvNeXt mapped into the same hub.
T_BGE	N x 1024	Target (true) BGE-M3 embeddings of the paired captions.
W_small	768 x 512	DINOv2-small entry map.
W_conv	1024 x 512	ConvNeXt entry map.
W_head	512 x 1024	Frozen caption head trained from H_small to T_BGE.
W_native	512 x 1024	ConvNeXt-specific comparison head trained from H_conv to T_BGE.

Models used in the hub experiment

Role	Models / dimensions	Contribution
Original hub sources	THREE configurations, and they are not interchangeable. FOUR spaces - DINOv2-small 768d, -base 1536d, -large 2048d, BGE-M3 1024d (5376d concatenated) - is the PUBLISHED protocol, and the configuration the 96.7 percent belongs to. FIVE spaces adds GPT-2 768d (6144d), Section C.13's body. SEVEN adds BERT and SBERT, 768d each (7680d), Section C.13.3.	Normalized vectors are concatenated, reduced to 512d by PCA/SVD, whitened, and frozen. The basis is estimated on the 8,533 TRAINING rows only.
Later visual encoder	ConvNeXt-base, 1024d; supervised ImageNet-22k pretraining.	It does not contribute to hub construction or W_head training. It only gets W_conv after arrival.
Caption target encoder	BGE-M3, 1024d.	Embeds the real paired captions. These vectors are the T in T_BGE: the training targets.

How The Frozen Hub Is Created

For every original encoder, run its model on the same paired items, normalize that encoder's columns, concatenate the resulting spaces, then reduce to a common whitened basis. ConvNeXt is absent from this process.

Order	Stage	Math	Cat/dog example	Project achievement
1	Collect original encoder vectors	X_k for each original encoder k	The cat image has one DINO-small vector, one DINO-base vector, etc. The caption "a cat on a chair" has a BGE vector.	Different vision and text encoders supply complementary views of the same paired items.
2	Normalize per encoder	$Z_k = (X_k - \mu_k) / \sigma_k$	No encoder gets extra influence simply because its activations have larger numeric scale.	Makes concatenated spaces comparable.
3	Concatenate	$Z = [Z_1 Z_2 \dots]$	The cat's visual and caption-related signals sit together in one wide row.	Creates one joint source matrix.
4	PCA/SVD and whiten	$H = \text{whiten}(\text{PCA}_{512}(Z))$	Cat and dog land in 512 shared coordinates; related cat captions tend toward the cat region.	A frozen 512d interoperability space, built before ConvNeXt appears.

The hub is not DINO-small. It is shared, multi-encoder geometry. DINO-small becomes special only because it is the sole input route used to train W_{head} .

What The 512 Dimensions Actually Are

A common and reasonable misreading: that PCA-512 SELECTS 512 vectors - for instance the 512 DINOv2-small items whose cosine similarity to something is highest. It does not. No item is chosen, ranked, or discarded, and no cosine similarity is computed anywhere in the hub's construction.

What PCA-512 does NOT do

- Pick 512 rows (items) out of 9,533.
- Pick 512 of DINOv2-small's 768 original dimensions.
- Rank anything by cosine similarity.
- Treat DINOv2-small as more important than the other sources.

What PCA-512 DOES do

- Builds 512 NEW axes, each a weighted combination of ALL 5,376 concatenated input dimensions.
- Orders those axes by how much variance across the 8,533 training rows each explains.
- Keeps the top 512 and discards the rest of the variance directions.
- Whitens: divides each kept axis by its singular value so all 512 have unit variance.

Why 512, and not 768 or 1,024. The width is bounded by the SOURCE encoder's algebraic rank, not chosen for accuracy. DINOv2-small outputs 768 dimensions, so at hub width 768 the last retained singular value is near-null (0.0054 against 10.79 one step earlier) and the least-squares fit for W_{head} divides by it - transfer collapses from 0.894 at $k=760$ to 0.009 at $k=768$. This is the width cliff of report Section E.8: a numerical artifact of the fit, not a property of the representations. The operating rule that follows is $\text{hub_width} < \text{dim}(\text{source encoder})$. 512 satisfies it with margin; 767 would be the widest safe value, and 512 was chosen well below the boundary rather than at it.

Where DINOv2-small IS special, and where it is not. It is one of several equal contributors to the hub basis - its columns are standardised exactly like every other source's, and PCA gives it no privileged weight. It becomes special only at the next stage: it is the sole encoder whose hub coordinates are used as the INPUT side when fitting W_{head} . That is a choice about the head's training route, made after the hub already exists, and nothing about the hub's 512 axes reflects it.

Training The Caption Head Exclusively Through DINOv2-Small

“Exclusively through DINOv2-small” describes the **input route used during fitting of W_head**. The head sees DINO-small’s hub coordinates and learns to predict the paired caption’s BGE vector. It does not receive ConvNeXt features, DINO-base features, or DINO-large features as head-training inputs.

Order	Stage	Math	Cat/dog example	Project achievement: how it was trained
5	Fit DINO-small entry map	$W_{\text{small}} = \text{ridge}(X_{\text{small,train}}, H_{\text{train}})$	The cat’s raw 768d DINO-small vector learns a linear route to its 512d hub location.	One ridge regression creates a DINO-small-to-hub adapter.
6	Create head inputs	$H_{\text{small}} = X_{\text{small}} W_{\text{small}}$	For cat image i: $h_{\text{small},i}$ is its DINO-small-derived hub coordinate.	The head gets 512d hub inputs, not raw 768d DINO vectors.
7	Fit caption head	$W_{\text{head}} = \text{ridge}(H_{\text{small,train}}, T_{\text{BGE,train}})$	Input: cat’s hub vector. Target: BGE vector for "a cat on a chair." The dog row uses its own matching caption.	The only training pairs for this head are $H_{\text{small}} \rightarrow T_{\text{BGE}}$.
8	Freeze head	$T\text{-hat} = H W_{\text{head}}$	The head has learned a linear rule that reads a hub location and produces a caption-semantic vector.	W_{head} is fixed before ConvNeXt transfer.

Ridge regression means the learned matrix minimizes squared error with a penalty on large weights:

$$\text{ridge}(A, B) = \text{argmin}_W \|AW - B\|_F^2 + \text{lambda}\|W\|_F^2. \text{ In closed form: } W = (A^T A + \text{lambda } I)^{-1} A^T B.$$

Here, **T** in T_{BGE} means **target**: the real BGE-M3 caption vector the prediction should match.

ConvNeXt Transfer: What Actually Runs

ConvNeXt is **not processed by DINOv2-small**. At test time, the image is passed through ConvNeXt, then ConvNeXt's own entry map, then the old DINO-trained head.

Order	Stage	Math	Cat/dog example	Project achievement
9	Cache ConvNeXt vectors	$X_{\text{conv}} \text{ in } \mathbb{R}^{N \times 1024}$	Run the cat and dog images forward once through ConvNeXt-base.	The one GPU feature-extraction step.
10	Fit ConvNeXt entry map	$W_{\text{conv}} = \text{ridge}(X_{\text{conv,train}}, H_{\text{train}})$	The cat's 1024d ConvNeXt vector learns to enter the existing 512d hub.	A small closed-form adapter; hub and head remain unchanged.
11	Map ConvNeXt into hub	$H_{\text{conv}} = X_{\text{conv}} W_{\text{conv}}$	The ConvNeXt cat becomes $h_{\text{conv,cat}}$, a compatible (not identical) hub coordinate to $h_{\text{small,cat}}$	Architecture-specific input becomes shared-space input.
12	Apply frozen old head	$T\text{-hat}_{\text{transfer}} = H_{\text{conv}} W_{\text{head}}$	The cat hub point goes into the DINO-trained reader and yields a predicted BGE vector near captions about the cat.	Tests whether the hub makes the same downstream function reusable across encoders.

Correct sentence: "ConvNeXt is mapped into the frozen hub, then read by a caption head trained on DINOv2-small hub coordinates."

Incorrect sentence: "ConvNeXt is processed by DINO-small in the hub." DINO-small is not on the runtime path. Its role is historical: it provided the hub-coordinate inputs that fitted W_{head} .

Retrieval, Native Comparison, And Meaning Of 96.7%

On each of the 1,000 held-out items, the predicted BGE vector queries a gallery of the 1,000 real BGE caption vectors. Cosine similarity ranks candidates. R@1 counts how often the paired caption is ranked first.

Method	Head fitting inputs	Prediction on ConvNeXt test item	Purpose
Transferred head	$H_{\text{small,train}} \rightarrow T_{\text{BGE,train}}$	$\hat{T}_{\text{transfer}} = H_{\text{conv,test}} W_{\text{head}}$	Measures reuse of a head trained exclusively through DINO-small.
Native ConvNeXt head	$H_{\text{conv,train}} \rightarrow T_{\text{BGE,train}}$	$\hat{T}_{\text{native}} = H_{\text{conv,test}} W_{\text{native}}$	A ConvNeXt-specific linear reference.
Random-entry control	Replace W_{conv} with a random matrix	Random hub-like input $\rightarrow W_{\text{head}}$	Checks that success is carried by the fitted entry map and hub rather than the scoring setup.

For the cat test image, the transfer prediction should rank the true BGE vector for “a cat on a chair” above the dog caption and the other 998 captions. A correct top rank contributes 1 to R@1; otherwise 0.

Reported relationship: $R@1_{\text{transfer}} / R@1_{\text{native}} = 96.7\%$. This says the transferred head recovers 96.7% of the native head’s achieved R@1. It does not say that transfer R@1 itself equals 96.7%.

Why this is notable

The head was trained via a self-supervised vision transformer, DINOv2-small (768d), and evaluated through a supervised convolutional network, ConvNeXt-base (1024d, ImageNet-22k). The architectures, feature dimensions, and training objectives differ. The bridge is the frozen 512d hub plus an independently fitted linear entry map. The stated ConvNeXt retention (96.7%) is also above the cited within-DINO-family band of 93.8%-96.5% for DINOv2-base/large transfer.

The random-entry control falling to 0.001 as a fraction (0.1%, the 1/1000 chance level) supports the interpretation that meaningful hub alignment, not the retrieval protocol itself, carries the transfer.

The Sharper Version Of The Same Point

ConvNeXt’s raw geometry agrees LEAST, yet it transfers BEST. Extending the shape-agreement matrix to eight encoders (report Section E.12) measures something the transfer number alone cannot: how much ConvNeXt’s raw pairwise-distance structure resembles each other encoder’s, before any hub or map exists. Spearman rank correlation over all 9,533 items, nothing fitted.

encoder	best raw agreement with text	hub transfer
DINOv2-large	$\rho = 0.418$ (with BERT)	92.9 percent of native
ConvNeXt-base	$\rho = 0.330$ (with SBERT)	96.7 percent of native

Why this matters for the claim in this document. ConvNeXt has the LOWEST raw shape agreement of any image encoder in the set - lower than every DINOv2 variant against text, and its agreement with the other IMAGE encoders (0.39 to 0.40) is well below what the three DINOv2 variants reach with one another (0.56 to 0.60). So the 96.7 percent is not underwritten by ConvNeXt’s geometry already happening to line up. It is achieved DESPITE the weakest raw agreement in the roster. The frozen hub and its one fitted entry map are recovering a linear correspondence that the raw-geometry measurement does not register at all - which is a stronger statement about the hub than the transfer number makes on its own.

One-Page Mental Model

Training route: cat image -> DINOv2-small -> X_small -> W_small -> H_small -> W_head -> predicted BGE caption. The real BGE vector for the cat caption is the supervision target.

Transferred test route: cat image -> ConvNeXt -> X_conv -> W_conv -> H_conv -> the same frozen W_head -> predicted BGE caption. No DINOv2-small forward pass occurs here.

Question tested: after each encoder is independently mapped into the frozen hub, can one downstream caption-reading rule learned from DINO-small still read ConvNeXt's hub coordinates?

Claim	Precise version
"The head never trained on ConvNeXt."	True. W_head is fitted only from H_small to T_BGE.
"ConvNeXt never trained in any form."	Too broad. W_conv is trained/fitted from X_conv to H; the caption head is not.
"ConvNeXt goes through DINO-small."	False. ConvNeXt goes through W_conv into the hub, then through the DINO-trained W_head.
"96.7% transfer."	Relative R@1 retention against the ConvNeXt-specific native linear head.
"All 9,533 were train rows."	Not for a clean 1,000-item held-out evaluation. Use 8,533 fitting rows and 1,000 test rows; disclose any transductive all-row PCA/map fitting.

Suggested project statement: "After ConvNeXt is independently aligned to the frozen hub with a linear entry map, a caption-retrieval head trained only on DINOv2-small hub coordinates retains 96.7% of the R@1 of a ConvNeXt-specific linear head. Under a strict split, all fitted transforms are trained on 8,533 items and retrieval is evaluated on 1,000 held-out items."